

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

22 Aug
260

CSA1407-Compiler Design

TEST! COA-PT1

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① If $(a < b)$ and $((c < d) \text{ or } (a > d))$ then

$$Z = x + y * z$$

else

$$Z = Z + 1$$

Soln!

$(a < b)$ and $((c < d) \text{ or } (a > d))$

Condition

$(a < b)$ (Three Address Code)
 $(c < d)$
 $(a > d)$

38/40
✓

L1:

If $a < b$ goto L2 # If condition
goto L5

L2: If $c < d$ goto L4
goto L3

L3: If $a > d$ goto L4
goto L5

L4:
 $t_1 = y * z$
 $t_2 = x + t_1$
 $z = t_2$
goto L6

L5:
 $t_3 = z + 1$
 $z = t_3$

L6:
goto L5

L1 = if $(a < b)$
L2 = and
L3 = $c < d$
L4 = or
L5 = $a > d$

② Translate $(a < b)$ and $(c < d)$ or $(e < f)$ using backpatching

Ans:

Backpatch statement $a < b$ to statement 3. $(e < f)$

1) IF $a < b$ goto 3

Backpatch statement $(c < d)$ to $(e < f)$

Backpatch statement 4 to statement 5

1) IF $a < b$ goto 3

2) goto 5

3) IF $c < d$ goto 7

4) goto 5

5) IF $e < f$ goto 7

6) goto 8

7) TRUE

8) FALSE

③ $A = (-A + B) * ((1 - A + B) * C)$ (TAC)
-A (UM)

Soln:

$t_1 = -A$

$t_2 = t_1 + B$

$t_3 = 1 - A$

$t_4 = t_3 + B$

$t_5 = t_4 * C$

$t_6 = t_2 * t_5$

$A = t_6$

Quadruples:

Operator	Arg1	Arg2	Result
uminus	A	-	t1
+	t1	B	t2
uminus	A	-	t3
+	t3	B	t4
*	t4	C	t5
*	t2	t5	t6
=	t6	-	A

④ Construct Three Address Code - $(a > b) \text{ or } (c < d) \text{ or } (e < f)$

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \&\& B2$

$B \rightarrow !B1$

$B \rightarrow E1 \text{ xlop } E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Soln!

L1: IF A > B goto LTRUE
goto L2

L2: IF C < D goto LTRUE
goto L3

L3: IF E < F goto LTRUE
goto LFALSE

LTRUE:

goto LEND // Boolean exp is True

LFALSE:

LEND: // Boolean exp is FALSE

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Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps